

Employee Payroll

1. Objective

To develop a Java program to demonstrate **method overriding and runtime polymorphism** by calculating and displaying the salary details of Regular and Contract Employees based on their employee ID.

2. Algorithm

1. Start the program.
2. Create a base class Employee with an employee ID.
3. Define a salaryDetails() method in the Employee class.
4. Create a RegularEmployee class that extends Employee.
5. Override the salaryDetails() method in RegularEmployee.
6. Set the basic pay, HRA, and T.A. for a regular employee.
7. Calculate the total salary.
8. Create a ContractEmployee class that extends Employee.
9. Override the salaryDetails() method in ContractEmployee.
10. Set the basic pay, HRA, and T.A. for a contract employee.
11. Calculate the total salary.
12. In the main() method, assign an employee ID.
13. Check whether the ID starts with "R" or "C".
14. Create the appropriate employee object using an Employee reference.
15. Call the overridden salaryDetails() method.
16. Display the salary details.
17. Stop the program.

3. UML Class Diagram

Employee - employeeId : String + Employee(String) + salaryDetails()

Extends

RegularEmployee + RegularEmployee(String) + salaryDetails()	ContractEmployee + ContractEmployee(String) + salaryDetails()
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EmployeePayroll + main(String[])

Employee Reference -> Regular / Contract Employee

4. Source Code

// Base class

```
class Employee{
```

```
    // Employee ID
```

```
    String employeeId;
```

```
    // Constructor
```

```
    Employee(String employeeId){
```

```
        this.employeeId = employeeId;
```

```
    }
```

```
    // Method to display salary details
```

```
    void salaryDetails(){
```

```
        System.out.println("Salary Details:");
```

```
    }
```

```
}
```

// Regular Employee class

```
class RegularEmployee extends Employee{
```

```
    // Constructor
```

```
RegularEmployee(String employeeId){
    super(employeeId);
}

// Overriding salaryDetails() method
@Override
void salaryDetails(){
    // Salary components for regular employee
    int basicPay = 25000;
    int hra = 15000;
    int ta = 5000;

    // Calculate total salary
    int total = basicPay + hra + ta;

    // Display salary details
    System.out.println("Salary Details:");
    System.out.println("Basic Pay: " + basicPay);
    System.out.println("HRA: " + hra);
    System.out.println("T.A: " + ta);
    System.out.println("Total Amount: " + total);
}
}

// Contract Employee class
class ContractEmployee extends Employee{
    // Constructor
    ContractEmployee(String employeeId){
        super(employeeId);
    }
}
```

```

// Overriding salaryDetails() method

@Override
void salaryDetails(){

    // Salary components for contract employee

    int basicPay = 12000;

    int hra = 0;

    int ta = 3000;

    // Calculate total salary

    int total = basicPay + hra + ta;

    // Display salary details

    System.out.println("Salary Details:");

    System.out.println("Basic Pay: " + basicPay);

    System.out.println("HRA: " + hra);

    System.out.println("T.A: " + ta);

    System.out.println("Total Amount: " + total);

}

}

// Main class

public class EmployeePayroll{

    public static void main(String args[]){

        // Employee ID input

        String employeeId = "R101";

        // Check employee type using ID

        if(employeeId.startsWith("R")){

            // Create RegularEmployee object

            Employee emp = new RegularEmployee(employeeId);

```

```
        // Calling overridden method
        emp.salaryDetails();
    }
    else if(employeeId.startsWith("C")){
        // Create ContractEmployee object
        Employee emp = new ContractEmployee(employeeId);
        // Calling overridden method
        emp.salaryDetails();
    }
    else{
        System.out.println("Invalid Employee ID");
    }
}
}
```

5. Output

Since the employee ID in the given program is "R101", the RegularEmployee object is created.

Salary Details:

Basic Pay: 25000

HRA: 15000

T.A: 5000

Total Amount: 45000

If the employee ID is changed to "C101", the output will be:

Salary Details:

Basic Pay: 12000

HRA: 0

T.A: 3000

Total Amount: 15000